

## 12 Chapter - 12 CSS Transition & Transform

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### → CSS Transition

- Transition makes a CSS property changes happen smoothly.

X Jab kisi element ki CSS property change hoti hai, transition us change ko smooth banata hai.

#### example

```
button {
```

```
background-color: blue;
```

```
transition: 0.5s;
```

```
}
```

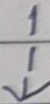
```
button:hover {
```

```
background-color: red;
```

```
}
```

#### Output:

Button  
Blue



Button  
red

X Button ka background color blue tha. jaise hi hamlog uspe jayenge to wo dhire se red me change hojaga (Smoothly)

### < 1 > transition-duration

- Defines how much time the transition takes.

X Change hone me kitna time lagega.

#### Example:

```
button {
```

```
transition-duration: 1s;
```

```
}
```

#### Output:

Hover  
transition start → 1 second  
change complete



12 Chapter-12 CSS Transition & Animation  
\_/\_/\_  
(ii) transition-property

- Defines which CSS property should change smoothly.

X Kaunsi CSS property smoothly change hogi ya define karta hai

Example:

.box {

transition-property: background-color;

transition-duration: 0.5s;  
}

.box: hover {

background-color: red;  
}

Output:

Before:

After:

Box  
Blue

Hover  
→

Box  
Red

(iii) transition-timing-function

- Controls the speed pattern of the transition

X Transition kis speed se start aur end hogi, ye control karta hai.

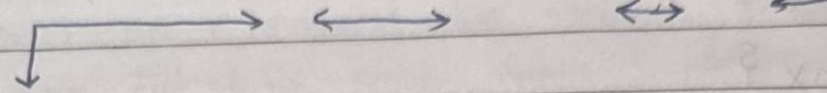


example:

```
.box {
```

```
  transition: transform 0.5s ease;
```

```
}
```



ease

→ Slow → Fast → Slow

linear

→ Same speed

ease-in

→ Slow start → Fast

ease-in-out

→ Slow start → Fast → Slow end

#### iv) transition-delay

- Adds a delay before the transition starts.

X Transition start hone se pehle wait karata hai.

example:

```
.box {
```

```
  transition: 0.5s;
```

```
  transition-delay: 1s;
```

```
}
```

Output:

Hover



1s wait



Transition starts



change happens.



## v> Transition Shorthand

Hum transition ki puri properties ko ek hi line mein likh sakte hain.

Before:

```
.box {
```

```
  transition-property: transform;
```

```
  transition-duration: 0.5s;
```

```
  transition-timing-function: ease;
```

```
}
```

After:

```
.box {
```

```
  transition: transform 0.5s ease;
```

```
}
```

↓  
property

↓  
Duration

↓  
Speed (timing)



## Transform

- Transform changes the position, size, angle on shape of an element.

X transform se element ko move, bada, chota, rotate ya tilt kar sakte hain.

Example:

translate() → Move

scale() → Size

rotate() → Rotate

skew() → Tilt

### i) translate() - Move

- Moves an element from its current position.

X Element ko ek position se doosri par move karta hai.

Example:

.box {

transform: translate(50px, 20px);

}

Output:

Before:

Box

After:

Box

→ 50px



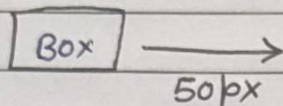
ii) translateX() & translateY()

- translateX()  
- Horizontal movement: (Ye left-right move karta hai)

example:

```
.box {  
  transform: translate(50px);  
}
```

Output:



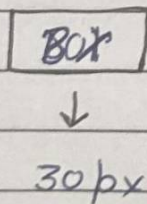
- translateY()

- Vertical movement: (Ye up/down move karta hai)

example:

```
.box {  
  transform: translateY(30px);  
}
```

Output:





### (iii) scale() - Change Size

- scale() makes an element bigger or smaller.
- X Element ka size increase ya decrease karne ke liye scale() use hota hai.

Example:

```
• box {
```

```
  transition transform: scale(1.2);  
}
```

### Meaning

scale(1) → Original size

scale(1.2) → 20% bigger

scale(2) → 2x bigger

scale(0.5) → Half size

### Output:

Before:

Box

After:

Box

Bigger



iv) rotate() - Rotate Element

- Rotates an element.

X Elements ko ghumane ke liye rotate() use karte hai.

example

```
.box {
```

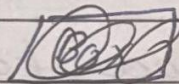
```
transform: rotate(45deg);
```

```
}
```

Output: Before:

Box

After:



Box

Common Values

45 deg - Slight rotation

90 deg - Quarter rotation

180 deg - Half rotation

360 deg - Full rotation.



### v) skew() - Tilt

- Tilts an element

X Element ko tedha ya tilt ke liye skew() use hota hai

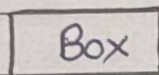
example:

.box {

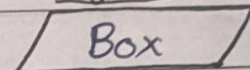
transform: skew(20deg);  
}

Output:

Before



After



### <vi> transform-origin

- Defines the point from which the transform happens.

X English

X Element kis point se rotate karana hai  
ya kis corner se ghumana hai

example:

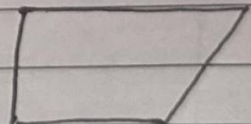
.box {

transform-origin: top left;  
transform: rotate(20deg);  
}

Values

center, top, bottom, left, right, top right,  
top left, bottom left, bottom right.

Output:





## → Practical

HTML

```
<button class="btn">Hover Me </button>
```

CSS

```
.btn {
```

```
padding: 10px 20px;
```

```
background: blue;
```

```
color: white;
```

```
transition: 0.3s;
```

```
}
```

```
.btn: hover {
```

```
background: red;
```

```
transform: scale(1.1);
```

```
}
```

Output

Normal

Hover

Blue

Hover

Hover

Red bigger



## → CSS CheatSheet

/\* Transition \*/

transition: transform 0.3s ease;

transition-property: transform;

transition-duration: 0.5s;

transition-timing-function: ease;

transition-delay: 1s;

/\* Transform \*/

/\* Move \*/

transform: translate(50px, 20px);

/\* Horizontal \*/

transform: translateX(50px);

/\* Vertical \*/

transform: translateY(30px);

/\* Size \*/

transform: scale(1.2);

/\* Rotate \*/

transform: rotate(45deg);

/\* Tilt \*/

transform: skew(20deg);

/\* Transform Point \*/

transform-origin: center;